

12.2 Classwork: Complex Numbers Restart (no calculator) — Markscheme

1. [Maximum mark: 7]

$$z = 3 - 2i, w = 1 + 4i.$$

- (a) Find zw . [2]

Solution:

$$(3 - 2i)(1 + 4i) = 3 + 12i - 2i - 8i^2 \quad \text{M1}$$

$$zw = 11 + 10i \quad \text{A1}$$

M1 for any complete expansion of the product, however the four terms are organised.

- (b) Find $\frac{z}{w}$ in the form $a + bi$. [2]

Solution:

$$\frac{3 - 2i}{1 + 4i} \cdot \frac{1 - 4i}{1 - 4i} = \frac{3 - 12i - 2i + 8i^2}{1 + 16} \quad \text{M1}$$

$$\frac{z}{w} = \frac{-5 - 14i}{17} = -\frac{5}{17} - \frac{14}{17}i \quad \text{A1}$$

M1 for any correct method of realising the denominator (multiplying by the conjugate, or equating $(a + bi)(1 + 4i) = 3 - 2i$).

Accept $\frac{-5 - 14i}{17}$ or any equivalent exact form. Do not accept a decimal approximation.

- (c) Find the exact value of $|z|$. [1]

Solution:

$$|z| = \sqrt{3^2 + (-2)^2} = \sqrt{13} \quad \text{A1}$$

- (d) Find real x, y with $(x + yi)(2 - i) = 7 + 4i$. [2]

Solution:

$$(x + yi)(2 - i) = 2x + y + (2y - x)i$$

$$\text{Equate real and imaginary parts: } 2x + y = 7 \text{ and } 2y - x = 4 \quad \text{M1}$$

$$x = 2, y = 3 \quad \text{A1}$$

M1 for any method that separates the equation into two real equations. A1 requires both values; the pair comes from one simultaneous system, so the two marks are not split.

Accept a solution obtained by dividing: $x + yi = \frac{7 + 4i}{2 - i}$, provided both values are correct.

2. [Maximum mark: 7]

$$z^2 - 4z + 13 = 0, z \in \mathbb{C}.$$

(a) Solve the equation. [3]

Solution:

$$(z - 2)^2 + 9 = 0 \quad \text{M1}$$

$$(z - 2)^2 = -9$$

$$z = 2 + 3i, z = 2 - 3i \quad \text{A1A1}$$

M1 for completing the square or for the quadratic formula; either route earns it. A1 for each root; accept $2 \pm 3i$ as a single expression.

(b) Explain why the two roots are complex conjugates. [1]

Solution:

The coefficients of the equation are real, so any non-real root occurs together with its conjugate. **R1**

R1 for any statement of the conjugate-root property, or for the equivalent observation that the roots differ only in the sign of the $\pm 3i$ term. A bare restatement of the two roots earns no mark.

(c) Sum and product of the roots of the equation with root $2 + i\sqrt{3}$. [2]

Solution:

The other root is $2 - i\sqrt{3}$.

$$\text{Sum} = (2 + i\sqrt{3}) + (2 - i\sqrt{3}) = 4 \quad \text{A1}$$

$$\text{Product} = 2^2 - (i\sqrt{3})^2 = 4 + 3 = 7 \quad \text{A1}$$

A1 for each of the sum and the product; the two are independent. The conjugate root must be used, stated or implied; a sum or product built from any other second root earns no mark.

(d) Write down such a quadratic equation. [1]

Solution:

$$z^2 - 4z + 7 = 0 \quad \text{A1}$$

Accept any non-zero real multiple, e.g. $2z^2 - 8z + 14 = 0$, and accept the factored form $(z - 2 - i\sqrt{3})(z - 2 + i\sqrt{3}) = 0$.

FT: award A1 for $z^2 - (\text{candidate's sum})z + (\text{candidate's product}) = 0$ formed correctly from part (c).